Design and Three-dimensional Printing of Eco-efficient Multimaterial Structures

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Abstract

Reducing the carbon footprint of advanced composites needs design strategies that unite structural performance with environmental considerations. Leveraging multimaterial 3D printing technologies could be vital in their fabrication if material, design and local fibre orientation could be reconciled. Here, we introduce a computational framework for multimaterial topology optimisation that minimises cradle-to-gate CO_2 emissions while preserving mechanical integrity. The method simultaneously optimises material distribution and fibre orientation, enabling additive manufacturing of eco-efficient fibre-reinforced composites. This strategy enables smooth transitions between fibre phases, while a robust optimisation algorithm ensures reliable convergence. To enable fabrication, a streamline-based toolpath generation method converts the optimised fibre orientations into an uninterrupted, printable path. We demonstrate the approach using carbon and bamboo fibre-reinforced polymers. The optimised designs redistribute carbon and bamboo fibres according to local stress demands, achieving significant reductions in environmental impact without compromising stiffness. Experimental validation through three point bending and full-field deformation mapping confirms strong agreement with numerical predictions. This integrated workflow bridges computational design with sustainable production for high-performance composites with a minimal environmental footprint.

Keywords: Topology optimisation, additive manufacturing, bamboo fibre, carbon fibre, eco-efficient

Introduction

Reducing the CO_2 footprint of high-performance materials has become a crucial challenge in modern structural and aerospace engineering. Lightweight composites, particularly carbon-fibre reinforced polymers (CFRPs), have transformed industries such as aerospace, automotive and renewable energy by offering unparalleled stiffness-to-weight ratios and superior fatigue performance. Yet, the environmental cost of such materials remains significant. The production of such materials involves energy-intensive precursors and complex manufacturing routes that lead to large cradle-to-gate CO_2 emissions [1, 2, 3, 4, 5, 6]. The need to achieve high mechanical performance with environmental responsibility drives the search for sustainable alternatives and hybrid approaches, linking performance-critical material use with renewable, low-impact reinforcements [7,

8]. These low-impact reinforcements defined here as materails possessing significantly lower cradle-to-gate embodied CO_2 and cumulative energy demand than traditional synthetic fibres.

A promising route is to combine high- and low-impact reinforcements within the same structure: keep carbon fibre only in the few regions where its exceptional performance is essential, and replace it elsewhere with renewable reinforcements such as bamboo-fibre composites. Bamboo grows rapidly, is widely distributed, and more sustainable than synthetic fibres, resulting in dramatically reduced embodied CO_2 [9, 10]. Although its stiffness and strength are lower than the composite material, bamboo exhibits favourable stiffness-per-CO_2 ratios [11], making it particularly advantageous when employed in regions of the structure that experience low stress. Spatially resolved material assignment [12, 13] offers a way to preserve global mechanical integrity while reducing environmental impact. This assignment is already a staple of industrial structural design for instance, in wind turbine blades where high-stiffness carbon-fibre spar caps are embedded within a glass-fibre or polymer shell [14]. However, while the concept of hybridisation is established, its application has historically been limited to relatively simple geometries or intuitive load paths. Instead of enforcing a uniform material throughout the structure, the optimisation algorithm evaluates each element and assigns the material that best matches the local stress, stiffness, or sustainability requirements across complex geometries and multi-axial loadings. This enables a graded, position-dependent material layout rather than a uniform material choice. Such hybrid strategies bridge the gap between mechanical performance and sustainability by allocating high-impact materials only to highly stressed regions and replacing them elsewhere with renewable alternatives. When combined with optimisation formulations that include environmental metrics, this enables simultaneous optimisation of stiffness and embodied CO_2 emissions in multi-material structures [15, 16, 17, 18].

Recent advances in multi-material topology optimisation make it possible to integrate environmental performance directly into structural design [19]. For example, traditional optimisation schemes have been extended to include life-cycle assessment (LCA) metrics such as embodied CO_2, cumulative energy demand, or environmental impact per unit stiffness (stiffness-normalised emission index) or per unit mass. This allows the designer to balance compliance and environmental cost within a unified mathematical framework. Recent studies have introduced multi-objective formulations that couple mechanical and environmental criteria, using stiffness-normalise indices, and demonstrate Pareto-optimal design that minimise both deformation and embodied greenhouse gas emissions (cradle-to-gate CO_2 equivalents) [20, 16, 15, 17]. Others have adopted data-driven or AI-assisted optimisation, linking material selection, geometry, and sustainability indicators [13, 21].

The practical engineering applications of multi-material optimisation requires consideration of manufacturing constraints and anisotropic material behaviour. In composite structures, mechanical performance depends critically on fibre orientation, layer sequence, and fibre continuity. Emerging methods in orientation-controlled additive manufacturing (AM) [22, 23] and automated fibre placement (AFP) have enabled direct translation of optimised, anisotropic material layouts into physical structures. Optimisation approaches now incorporate orientation filtering, minimum feature control, and print-path continuity, ensuring that the optimised fibre paths are manufacturable [24]. In practice, this means that the optimised fibre trajectories

must not only satisfy theoretical stress-flow patterns but also comply with kinematic limits of deposition tools, curvature constraints, and interlayer bonding requirements. As these manufacturing aware algorithms evolve, this emulates the adaptive and efficient-driven growth patterns observed in nature [25, 26].

In natural load-bearing materials such as wood, structural optimisation arises from fibre alignment along principal stress trajectories, producing lightweight yet robust architectures through efficient material placement rather than high-density matter. Beyond orientational optimisation, biological systems enhance performance by combining constituents with distinct mechanical roles. Translating these motifs into synthetic systems, researchers have demonstrated that brick-and-mortar architectures and mineral nano-interconnectivity can simultaneously enhance stiffness, strength, and damage tolerance compared to brittle synthetic composites [27, 28].These examples highlight two principles directly relevant to engineered composites. First principle is spatially varying material placement and orientation according to local mechanical demands, and secondly, functional hybridisation of dissimilar constituents to achieve properties unattainable by a single material. Importantly, these biological strategies align with the emerging direction of multi-material topology optimisation, where material type, volume and orientation are treated as design variables rather than fixed inputs. In the context of sustainable engineering, this paradigm enables high-impact materials to be concentrated only where structurally necessary, while renewable, low-impact reinforcements occupy low-stress regions. While nature has served as a conceptual bridge for structural engineering for decades, historical applications often lacked the refinement necessary to capture the multi-scale, multi-objective complexity of biological systems. However, translating these principles into practice requires computational tools capable of handling material incompatibility, anisotropy, manufacturability and environmental metrics within a unified framework, motivating the formulation developed here.

Despite the recent progress, unifying mechanical and environmental objectives in a computationally efficient approach remains an open challenge. In structural topology optimisation, the typical foci are on compliance or weight minimisation, while in the eco-efficient framework, emphasis lies on embodied energy, cradle-to-gate-CO_2 emissions, or life-cycle impact minimisation. Integrating these objectives requires a multi-objective optimisation formulation capable of expressing trade-offs between mechanical efficiency and environmental cost [19, 23]. Weighted-sum formulations can lead to dominance of one objective over the other, while Pareto-front or normalisation approaches demand high computational cost and careful scaling of heterogeneous metrics. Moreover, maintaining numerical stability and manufacturability adds further complexity, particularly when discrete material transitions and anisotropic behaviour are involved. Reliable coupling of LCA data with material properties is essential, yet current databases often lack consistent links between mechanical properties and environmental indicators [7, 8]. This disconnect prevents the direct integration of sustainability data into the optimisation framework. Therefore, a robust framework needs three components: normalised and compatible objective functions for structural and environmental performance, scalable algorithms that preserve convergence and manufacturability, and a unified material dataset that quantifies both mechanical and environmental parameters. Only through such integration topology optimisation could evolve from

a purely geometric tool into a quantitative decision-making framework for low-carbon, high-performance structures.

Here, we present a bamboo-carbon fibre multi-material topology optimisation formulation that explicitly minimises the structure's CO_2 footprint while maintaining stiffness and weight targets (Figure 1). The formulation combines compliance and stiffness-normalised CO_2 emissions within a density-based multi-material scheme. The design domain is parametrised using separate density fields for each fibre-reinforced composite and an orientation field to capture anisotropy.

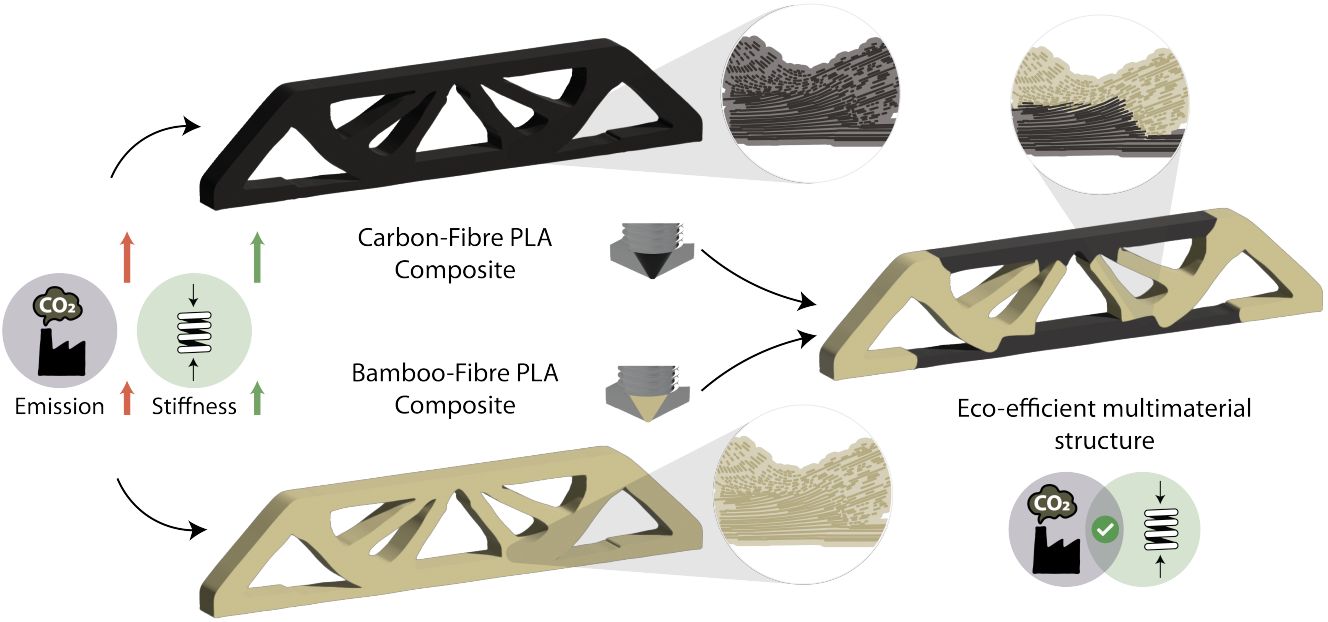

Figure 1: Schematic of our approach, shows that by incorporating mechanical and environmental factors of carbon-fibre reinforced Polylactic acid (PLA) composite (CF-PLA) and bamboo-fibre PLA composite (Bamboo-PLA), we propose a multi-material, eco-efficient, sustainable structures. The framework integrates a combination of compliance and CO_2 emissions data into a multi-objective topology optimisation. Additive manufacturing constraints, including fibre orientation control, minimum feature size, and direct print-path generation, ensure the optimised designs are practical to produce while retaining continuous fibre reinforcement.

By employing the uniform multi-material interpolation (UMMI-2) method [29] together with convolution and Heaviside filters [30, 31], the approach achieves smooth yet distinct material boundaries and continuous fibre trajectories. The environmental impact of each phase is evaluated using cradle-to-gate CO_2 intensity data [32, 20], enabling a fair comparison between renewable and synthetic fibres on a per-stiffness basis. The optimisation problem is solved using the Method of Moving Asymptotes (MMA) [33] with adjoint sensitivity analysis for efficient convergence. Finally, additive manufacturing constraints such as fibre orientation continuity, feature-size limits, and print-path generation are applied as a post-processing step to ensure that the optimised layouts are physically realisable. We demonstrate that the targeted replacement of CF-PLA with Bamboo-PLA, guided by environmental data and manufacturable anisotropic optimisation, can achieve substantial reductions in embodied emissions with minimal loss of stiffness, marking a step toward sustainable high-performance composites.

The key contribution of this work is the development of a computational framework that couples multi-material topology optimisation with environmental performance metrics, enabling designers to explicitly trade off stiffness against embodied CO_2 emissions according to application-specific requirements. The formulation integrates anisotropic material modelling,

fibre-orientation optimisation, and cradle-to-gate LCA data within a unified density-based scheme. Beyond simulation, we demonstrate the physical realisation of the optimised layouts through multimaterial, fibre-aligned 3D printing of carbon- and bamboo-reinforced composites. Taken together, this provides a practical pathway for designing sustainable, high-performance composite structures where mechanical efficiency and environmental impact can be balanced in a rational and application-driven manner.

Results and Discussion

First, we formulate the eco-design framework that simultaneously optimise the geometry of the structure, reinforcement directions and the multimaterial distribution for fibre-reinforced composites. In this work, we focus on structures made of two fibre-reinforced composites.

Design parameterisation

Considering two fibre-reinforced composite materials, we describe a design using three design fields, two density fields for three phases (two material phases and one void phase) and one fibre-orientation field. We take piecewise constant parametrisation for these design fields based on a finite element mesh. Therefore, there would be $3n$ design variables for a discretized geometry of n finite elements. The vector of design variables can be given as $\mathbf{X} = [\boldsymbol{\rho}^1, \boldsymbol{\rho}^2, \boldsymbol{\Theta}]$. Here, $\boldsymbol{\rho}^i = [\rho^i_1\ \rho^i_2\ ...\ \rho^i_j\ ...\ \rho^i_n],\ m = 1, 2$, with ρ^i_j being the variable associated with the i^{th} density field and the j^{th} element, while $\boldsymbol{\Theta} = [\theta_1\ \theta_2\ ...\ \theta_j\ ...\ \theta_n]$ with θ_j being the variable associated with the fibre orientation of the j^{th} element with respect to the x-axis.

Optimisation problem

Next, we formulate the optimization problem. The objective is to minimise a normalised CO_2 footprint of fibre-reinforced composite structure [20], while ensuring sufficient mechanical performance with a limit on maximum volume usage. The corresponding density topology optimisation problem can be defined as,

$$\min_{\mathbf{X}\in\mathbb{R}^{3n}} \frac{CO_2(\mathbf{X})}{CO_{2,0}}, \tag{1}$$

such that,

$$\frac{C(\mathbf{X})}{C_0} - 1 \leq 0 \quad \text{with} \quad C(\mathbf{X}) = \frac{1}{2}\mathbf{U}^T\mathbf{K}\mathbf{U}, \tag{2a}$$

$$\frac{V(\mathbf{X})}{V_0\, f} - 1 \leq 0 \quad \text{with} \quad V(\mathbf{X}) = \sum_{e=1}^{n}(\rho^1_e + \rho^2_e)v_e, \tag{2b}$$

$$0 \leq \rho^1_e \leq 1 \quad \text{with} \quad e = 1, 2, ..., n, \tag{2c}$$

$$0 \leq \rho^2_e \leq 1 \quad \text{with} \quad e = 1, 2, ..., n, \tag{2d}$$

$$-\pi \leq \theta_e \leq \pi \quad \text{with} \quad e = 1, 2, ..., n, \tag{2e}$$

and

$$\mathbf{K}\mathbf{U} = \mathbf{F} \tag{3}$$

where $CO_2(\mathbf{X})$, $C(\mathbf{X})$ and $V(\mathbf{X})$ are the CO_2-footprint, compliance and volume of the structure; $CO_{2,0}$, C_0 and V_0 is the CO_2-footprint, compliance and volume are the reference values used for normalisation; V_e denotes the elemental volume. f is the maximum allowed volume fraction; $\mathbf{K}$, $\mathbf{F}$ and $\mathbf{U}$ are the global stiffness matrix, the global force vector, and the global displacement vector, calculated using a finite element framework, respectively.

Here, Eq.(2a) is the constraint on compliance to ensure the sufficient stiffness of the structure; Eq.(2b) is the constraint on the volume fraction, this constraint can be easily substituted with an equivalent mass constraint; Eq.(2c), Eq.(2d) & Eq.(2e) are the box constraints on the design variables; and Eq.(3) is the state problem under consideration. Please note that the state problem is not applied as an explicit constraint in optimization, but is solved in each iteration to calculate the state variables, $\mathbf{U}$. To fully specify the optimization problem, we define crucial numerical components of the framework, filtering, material interpolation, and environmental impact in the following sections. To ensure manufacturable and mesh-independent designs, convolution-based filters are applied to the density and orientation fields[Eqs. (6)–(8), followed by a Heaviside projection that promotes crisp black-and-white layouts Eq. (9). The filtering method regularises the orientation field, avoiding sharp fibre curvatures that would otherwise hinder manufacturability and physical consistency.

Material Interpolation Scheme

A material interpolation scheme is crucial since it links the geometric design description to the corresponding material distribution and then eventually to the state problem. The interpolation scheme further incorporates a power-law density penalization to discourage intermediate density values and reduce the occurrence of large gray zones. For a single material optimisation, the Solid Isotropic Material Interpolation (SIMP) or Rational Approximation Material Interpolation (RAMP) is the commonly used material interpolation scheme. However, these schemes could be extended in more than one way for a multimaterial case, such as Recursive multiphase materials interpolation (RMMI), Uniform multiphase materials interpolation (UMMI), NURBS-based multi-material interpolation (NMMI) [34, 35, 29, 36, 37]. In our work, we use the UMMI-2 approach proposed in Gao *et al.* [29] due to its ability to obtain a superior optimum for uniform initial design variables and its straightforward implementation in density topology optimisation. As it requires no auxiliary variables or coupling constraints beyond the standard SIMP framework [29]. Using the UMMI-2 approach, the elemental-elasticity matrix $\mathbf{C}_e$ can be written as:

$$\mathbf{C}_e(\hat{\rho}_e^1, \hat{\rho}_e^2, \tilde{\theta}_e) = \mathbf{C}_{\text{void}} + (\hat{\rho}_e^1)^p \, (1 - (\hat{\rho}_e^2)^p) \left(\mathbf{R}(\tilde{\theta}_e)^T \mathbf{C}_1 \mathbf{R}(\tilde{\theta}_e)\right) + (1 - (\hat{\rho}_e^1)^p) \, (\hat{\rho}_e^2)^p \left(\mathbf{R}(\tilde{\theta}_e)^T \mathbf{C}_2 \mathbf{R}(\tilde{\theta}_e)\right) \tag{4}$$

where $\mathbf{C}_{\text{void}}$, $\mathbf{C}_1$ and $\mathbf{C}_2$ are the elasticity matrices for the void phase, material phase 1 and material phase 2, respectively. $\mathbf{R}$ is the local 2D rotation matrix to rotate the transversely isotropic elasticity matrix according to the filtered fibre-orientation $\tilde{\theta}_e$. p is the penalisation exponent used in the SIMP-type material interpolation to reduce intermediate density values and promote discrete material distributions. And, β and η parameter control the steepness and the threshold of the Heaviside function.

Environmental impact evaluation

The environmental impact of a structure (as in Eq. (14)) can be evaluated based on various quantifiable impact so called "mid-point" categories [20, 32] such as the green house gas emmisions (GHG), the acidification , the ozone depletion , the eutrophication , etc. In this work, we quantify the environmental cost using GHG emissions, focusing on the cradle-to-gate phase. Specifically, our system boundary encompasses raw material extraction and the primary processing required to produce the composite feedstock which means terminating at the point the material is ready for fabrication. By setting the 'gate' at the material level, we treat the emissions associated with specific manufacturing processes (e.g., additive manufacturing energy) as a constant overhead excluded from the optimisation objective, allowing the framework to focus on the trade-off between mechanical performance and the embodied impact of the constituent materials. Consequently, the environmental impact is defined as,

$$CO_2(\mathbf{X}) = \sum_{e=1}^{n} \left(\hat{\rho}_e^1 \, CO_2^1 + \hat{\rho}_e^2 \, CO_2^2 \right) v_e, \tag{5}$$

where CO_2^1 and CO_2^2 are the CO_2 footprint associated with the unit volume of material phase 1 and material phase 2, respectively. These values were obtained from Idemat and Ecoinvent, and eco-costs midpoint tables [38]. The extension of this formulation to an arbitrary number of materials is provided in the Materials and Methods section.

Optimizing a Messerschmitt - Bölkow - Blohm beam

In this subsection, we employ the proposed eco-design framework to design a Messerschmitt - Bölkow - Blohm (MBB) beam made of carbon fibre-reinforced PLA (CF-PLA) and bamboo fibre-reinforced PLA (Bamboo-PLA). The optimisation problem is solved using the Method of Moving Asymptotes (MMA). To calculate the required sensitivities, we use the commonly used adjoint method. Figure 2a presents the design domain with the boundary conditions. Figure 2b represents flow chart of the proposed optimization framework which the initial values for ρ_e^1, ρ_e^2 and θ_e are $f/2$, $f/2$ and 0 for all element e respectively. The material properties corresponding to each composite phase are listed in Table 1. Specifically, Figure 2c tracks the evolution of the CO_2 footprint (orange), compliance (teal), and total volume fraction (purple). Horizontal dashed lines represent the upper limits for compliance and V_f, while vertical dashed lines denote continuation steps where p and β increase. Figure 2d shows the evolution of the material volume fractions for both materials, revealing the optimiser's strategy of progressively reallocating CF-PLA and Bamboo-PLA to achieve both mechanical and environmental objectives.

This figure highlight that the design framework substantially reduces the CO_2-footprint of the structure while ensuring limits on mechanical performance and volume usage. Although each continuation step introduces a sudden jump in the functional values, the design evolves smoothly between these steps. The sudden jumps reflect the sudden change to the material interpolation scheme due to stricter penalization and to discreteness of the design due to steeper Heaviside projection. These sudden jumps may lead to temporary constraint violations, which are resolved as the solution stabilizes in subsequent iterations. The

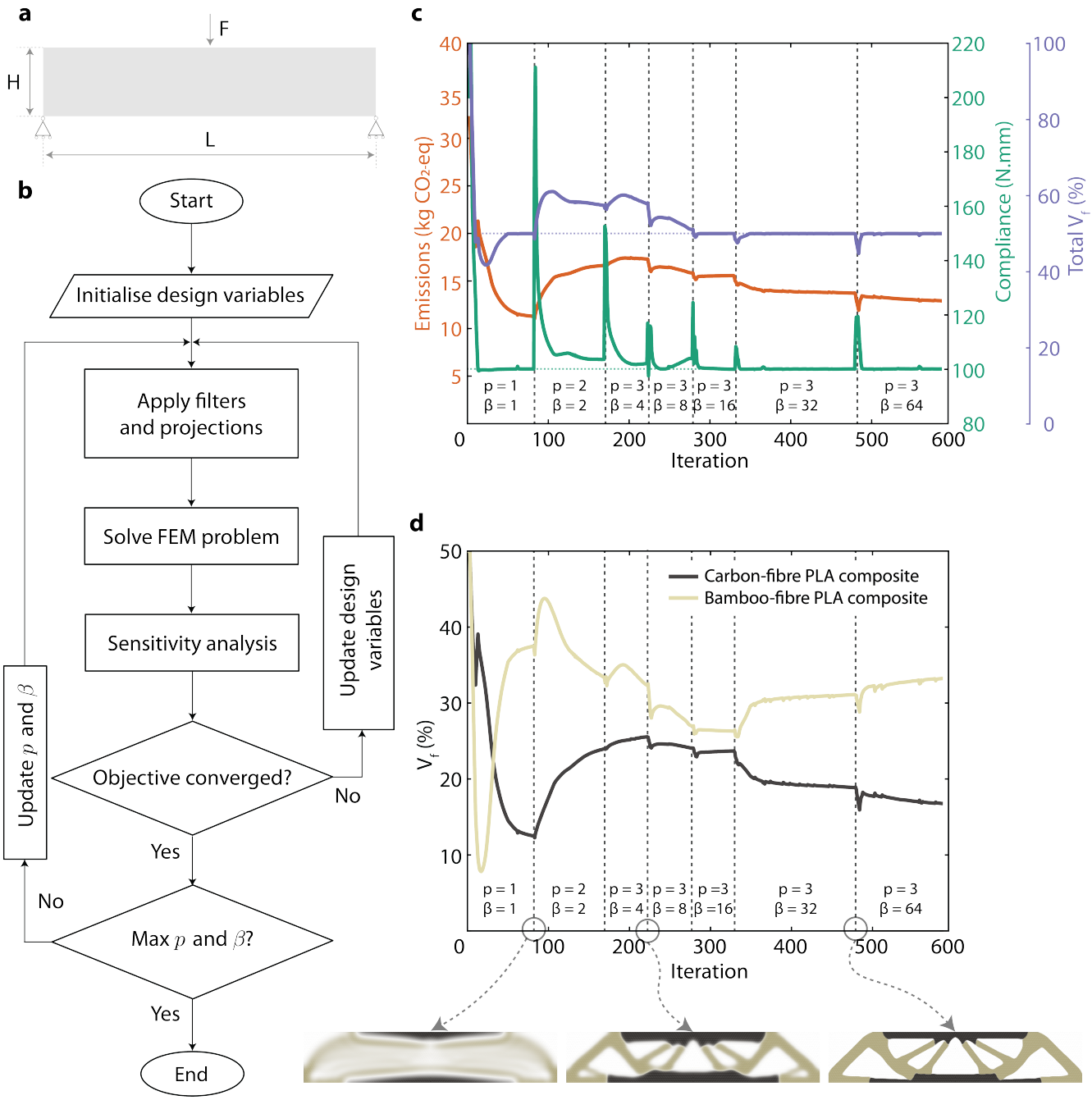

Figure 2: a) Schematic of the MBB beam with applied load and boundary conditions as F: loading, H: height = 40 mm, and L: the length of the beam = 200 mm. b) Workflow of the proposed multimaterial topology optimisation framework coupling density-based design for carbon- and bamboo-fibre phases with fibre-orientation control. The formulation minimises cradle-to-gate CO_2 emissions under compliance and volume constraints, employing convolution and Heaviside filters for manufacturability, and UMMI-2 for material interpolation [Eq. (11)]. c) Convergence history of compliance, CO_2 objective, and volume fraction; dashed lines denote constraint thresholds and continuation steps. d) Optimised material layouts: CF retained along critical load paths, BF redistributed to low-stress regions, achieving reduced environmental impact without compromising structural performance.

impact of these violations diminishes in the later stages of the continuation scheme compared with the earlier stages.

Figure 2d also presents the designs at iterations 100, 200, and 500. From these designs, we observe that the initial relaxed stages, with low p and β, allow the optimiser to explore a larger design space dominated by intermediate densities and avoid poor local minima. In the early iterations, the penalisation and projection parameters are still relaxed, allowing the optimiser to explore a broad design space with many intermediate densities. In this regime, the structure tends to spread stiffness away from the neutral axis and does not yet commit to discrete load-bearing members. This phase essentially allows the algorithm to freely identify potential load paths without over-penalising intermediate material usage. As the continuation scheme progresses, the penalisation and projection steepness increase, driving the design toward more discrete material layouts. At this stage, a clear division of roles emerges between the two composites: Bamboo-PLA begins to populate the low-stress regions, while CF-PLA increasingly concentrates along the primary load paths where stiffness is essential.

In the final iterations, the optimiser converges to a stable, fully feasible solution with a highly efficient hybrid configuration: CF-PLA is used only along structural members having a major stiffnes contribution, while Bamboo-PLA fills the remaining volume. This arrangement significantly reduces environmental impact without compromising stiffness. Overall, the material layout transitions from an initial "sandwich-like" distribution, where CF-PLA forms outer skins and Bamboo-PLA fills a diffuse core, to a final "truss-like" topology with discrete material domains that reflect true load-carrying paths.

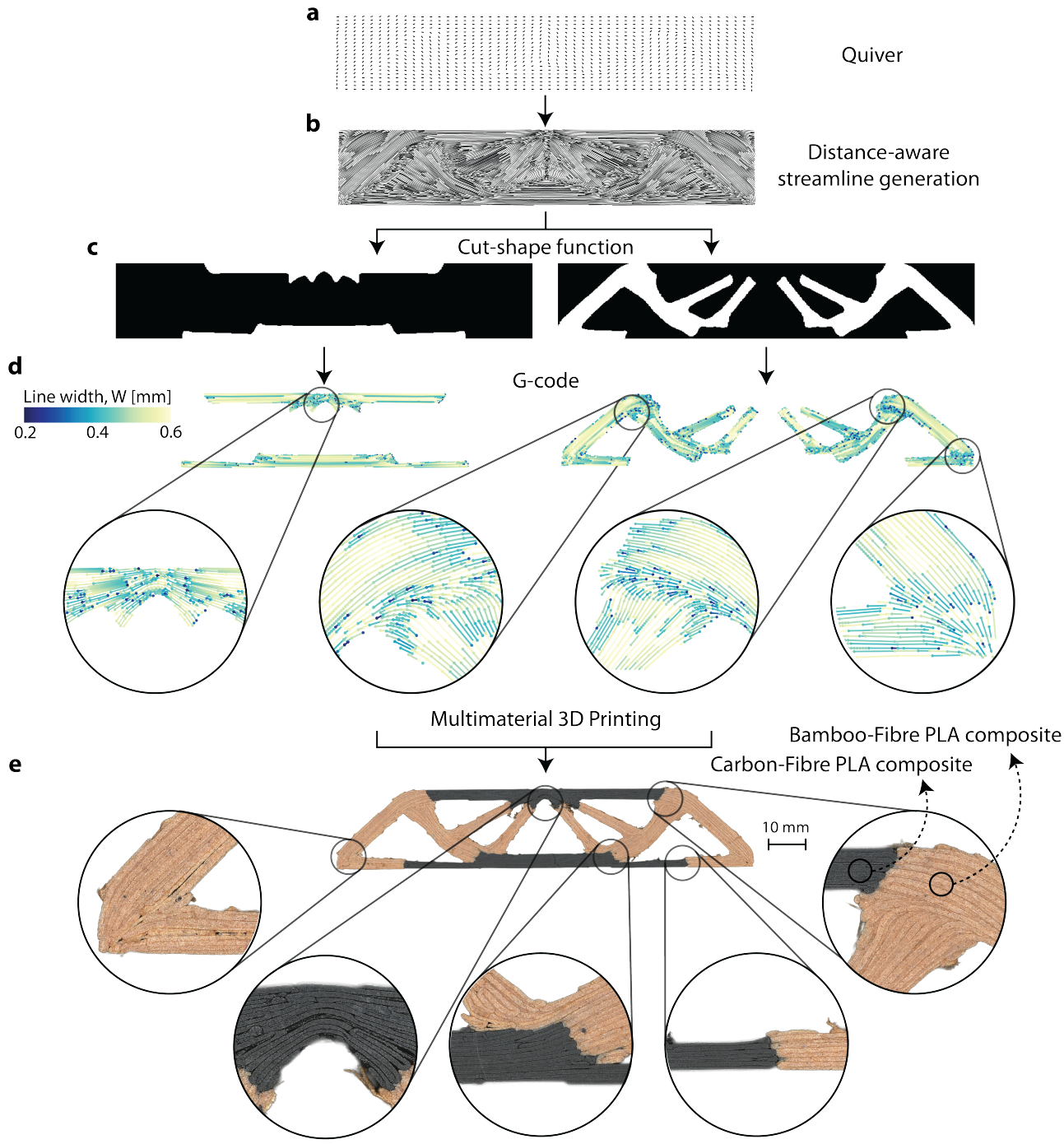

Figure 3: From optimised fibre orientations to multimaterial printing. a) The fibre orientation field from the topology optimisation is shown as quiver plots, indicating principal stress-aligned directions. b) These fields are converted into distance-aware streamlines, ensuring ensuring continuous and non-intersecting toolpaths with manufacturable spacing. c) The material density is converted into binary indicator functions that segment the domain into Bamboo-PLA and CF-PLA regions. Toolpaths are clipped accordingly and assigned variable extrusion widths. d) The resulting G-codes were merged and printed on a multimaterial printer with two dedicated print heads. e. The insets highlight the precise line width adjusted filling of the domain and smooth transitions between materials, confirming strong agreement between the designed fibre paths and the printed structure.

Printing of the optimised Messerschmitt - Bölkow - Blohm beam

The optimiser outputs element-wise density distribution and fibre-orientation distribution of the final design. These results are the digital blueprint for the optimised design. To physically realise this intricate digital result, we design a computational pipeline for flow-inspired anisotropic printing based on Houriet *et al.* [22]. First, the optimised fibre-orientation field is translated into a network of distance-aware streamlines. In line with Houriet's approach [22], streamlines are placed such that the maximum distance between neighbouring paths does not exceed a prescribed manufacturable bound, resulting in continuous, non-overlapping coverage of the domain with curvature-controlled trajectories. This ensures that the anisotropic reinforcement encoded by the optimiser is preserved while respecting geometric deposition constraints associated with fused-filament fabrication. This step is crucial to accurately capture the directional reinforcement behaviour predicted by topology optimisation while maintaining manufacturability. The algorithm dynamically adapts local path spacing based on curvature and stress-field gradients, mirroring how biological materials, such as wood, organise fibres around knots or defects to manage stress concentrations [22].

To fabricate the multimaterial design, the optimiser's material allocation is used to separate the field-aligned streamlines into CF-PLA and Bamboo-PLA toolpaths. Each set is clipped to its designated material region to ensure that no paths cross into the other material. This produces two independent toolpath sets with locally varying line widths to control extrusion volume. The two toolpaths are then merged into a single multimaterial build sequence with explicit extruder assignments.

The multimaterial structures were fabricated on a Prusa XL system equipped with independent toolheads, enabling clean transitions between short fibre CF-PLA and Bamboo-PLA. Both filaments are dried prior to printing to ensure stable extrusion. The parts are produced using a 0.2 mm layer height and standardised thermal settings, with automated retraction and purging to avoid cross-contamination during material switching. As shown in the magnified views of Figure 3, the printed filament trajectories reproduce the distance-aware streamline field with high fidelity, densely filling the domain and preserving continuous, curvature-controlled paths. The close match between the deposited filaments and the optimised fibre-orientation field confirms the robustness of the field-aligned toolpath generation and its ability to translate anisotropic design intent into a physical composite architecture. This demonstrates a fully integrated workflow in which multimaterial topology optimisation, orientation-aware path planning, and multi-nozzle additive manufacturing operate to realise sustainable, high-performance structures. To verify that this manufacturing translation also preserves the intended mechanical behaviour, we next experimentally characterise the printed beams under bending loads representative of their design case. Specifically, we use three-point bending to quantify the effective bending stiffness across design variants and to assess how the carbon–bamboo material allocation influences the stiffness–CO_2 trade-off.

Testing of the optimised Messerschmitt - Bölkow - Blohm beam

Here, we experimentally validate the optimised MBB beam using the three-point bending test (Figure SI1). Figure 4 provides the validation results for five design variants, labelled Design 1 through Design 5. Design-1 and Design-5 correspond to a fully Bamboo-PLA structure and a fully CF-PLA structure, respectively. Designs 2-4 are obtained using the optimisation framework with different compliance limits. From Designs 2 to 4, the CF-PLA content progressively increases, while the Bamboo-PLA content progressively decreases. In Figure 4a, the effective bending stiffness measured from three-point bending tests is plotted (17 and 19). The experimental stiffness increases monotonically with the CF content, demonstrating that the global mechanical response is governed primarily by CF-PLA. The printed samples below the plot, coloured by their calculated CO_2 footprint, highlight the relative difference in CO_2-footprint. The results of the CO_2 footprint and bending stiffness reveal the trade-off between structural stiffness and environmental impact attained by the optimizer.

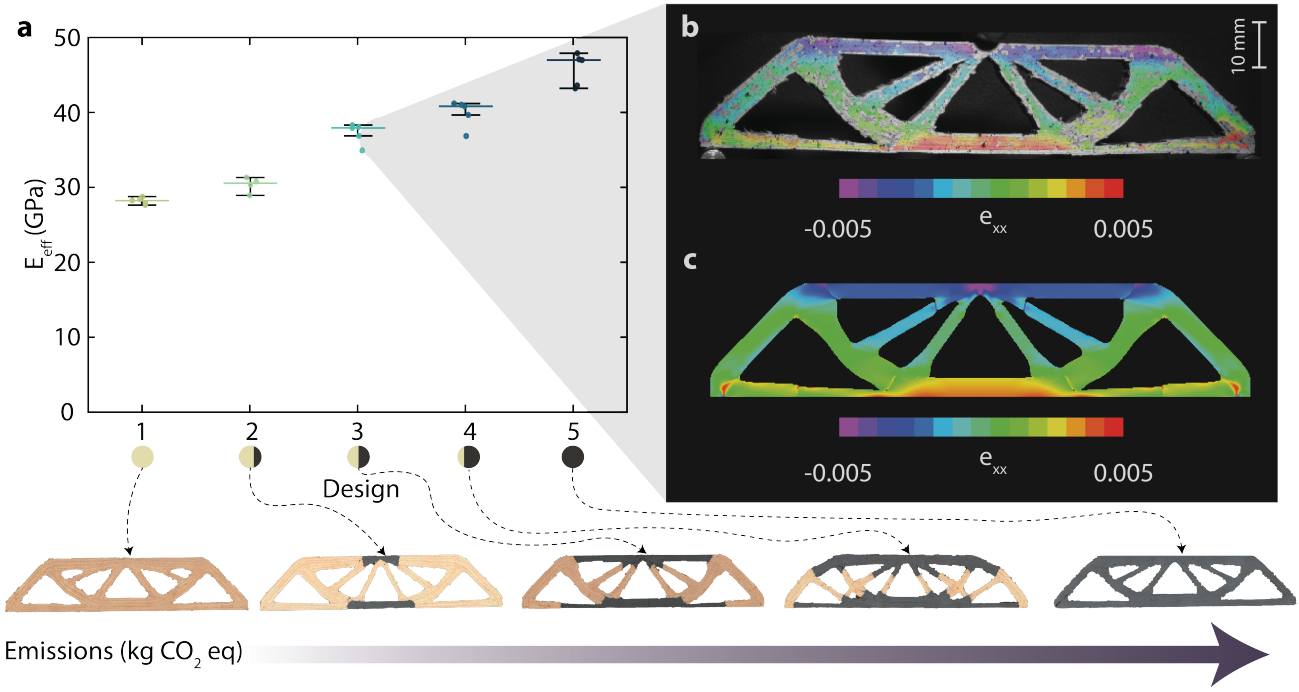

Figure 4: a) Three-point bending stiffness of five beam designs spanning the full range of CF-PLA volume fractions, from fully Bamboo-PLA (Design 1) to fully CF-PLA (Design 5). The corresponding printed beams are shown beneath, shaded by their embodied CO_2 emissions to illustrate the coupled stiffness–emissions gradient. Together, these results demonstrate that stiffness increases monotonically with carbon content while emissions rise accordingly. b) Digital image correlation (DIC) maps of the printed multimaterial structure under loading, showing full-field strain distributions for e_{xx}. c) Corresponding finite element analysis (FEA) strain predictions, demonstrating excellent agreement with experimental measurements. The combined results validate the entire workflow, from topology optimisation and material allocation to fibre-aligned printing and mechanical performance.

Design 3, which offers a balanced compromise between stiffness and environmental impact, was selected for detailed deformation analysis. Figure 4b shows the DIC-measured longitudinal strain field e_{xx} under bending. The tensile strains along the lower chord, compressive strains along the upper chord, and smooth gradients within the diagonal members confirm that the multimaterial layout maintains structural continuity and effective load transfer, without premature localisation at material

interfaces. The corresponding finite element prediction in Figure 4c reproduces these features with high fidelity. Minor deviations near the interfaces are attributed to local printing heterogeneity, but the overall agreement validates the constitutive modelling, the topology-optimised multimaterial layout, and the fibre-aligned toolpaths. Together, these results confirm that the computational design reliably translates into the structural response of the printed multimaterial structure.

Pareto front analyses

Figure 5 synthesises two complementary perspectives on the mechanical and environmental performance of the optimised multimaterial beams: a compliance-volume fraction Pareto analysis and a constraint-driven exploration of displacement and mass limits. In Figure 5a, the Pareto front is obtained by first computing the single-material compliance bounds for each prescribed total volume fraction V_f. Compliance minimisation with pure CF-PLA produces the lower bound, while pure Bamboo-PLA defines the upper bound. Three intermediate compliance targets are then imposed as constraints for the multimaterial optimisation, where the objective is minimisation of the cradle-to-gate CO_2 footprint. The resulting hybrid designs populate the interior of the Pareto surface, revealing how CF-PLA is introduced only where mechanically essential and how substantial emission reductions can be achieved relative to fully carbon structures for modest increases in compliance. However, since compliance is a global and somewhat abstract constraint, the Pareto view in Figure 5a does not directly reveal how the designs evolve under more application-driven requirements such as explicit deformation limits and strict weight budgets; this motivates the complementary analysis in Figure 5b.

Building on the Pareto trade-off in Figure 5a, Figure 5b reformulates the optimisation problem using directly prescribed engineering constraints, namely limits on maximum displacement and total mass (rather than an explicit compliance bound). Figure 5b presents the achievable design space when both maximum displacement ($u_{\max} = 0.8, 0.9, 1.0$ mm) and total mass (0.03-0.05 kg in 0.005 kg steps) are varied simultaneously. Each point represents a CO_2-minimising multimaterial topology with the imposed constraints. At low allowable mass, the displacement constraint cannot be fully maintained and the optimiser retains isolated CF-PLA regions to preserve stiffness, resulting in higher emissions. As the mass increases, the optimiser gains flexibility to reduce carbon usage while still meeting the displacement limit, leading to lower CO_2 values. Beyond roughly 0.045-0.050 kg, the designs reach a plateau where additional mass no longer decreases emissions, either because the displacement constraint is already comfortably satisfied or because the optimiser converges to a local minimum reinforced by the high projection steepness.

Finally, Figure 5c compiles the catalogue of multimaterial topologies produced across volume fractions from $V_f = 30\%$ to 70%. Low-volume designs form slender, bamboo-dominated truss-like layouts, whereas higher-volume designs develop more continuous load paths with CF-PLA concentrated in regions of high stress. This systematic progression provides a practical design map, offering a continuum of manufacturable configurations that balance displacement limits, mass budgets, and environmental impact within a unified optimisation framework. It is important to recognise that the shape of this design

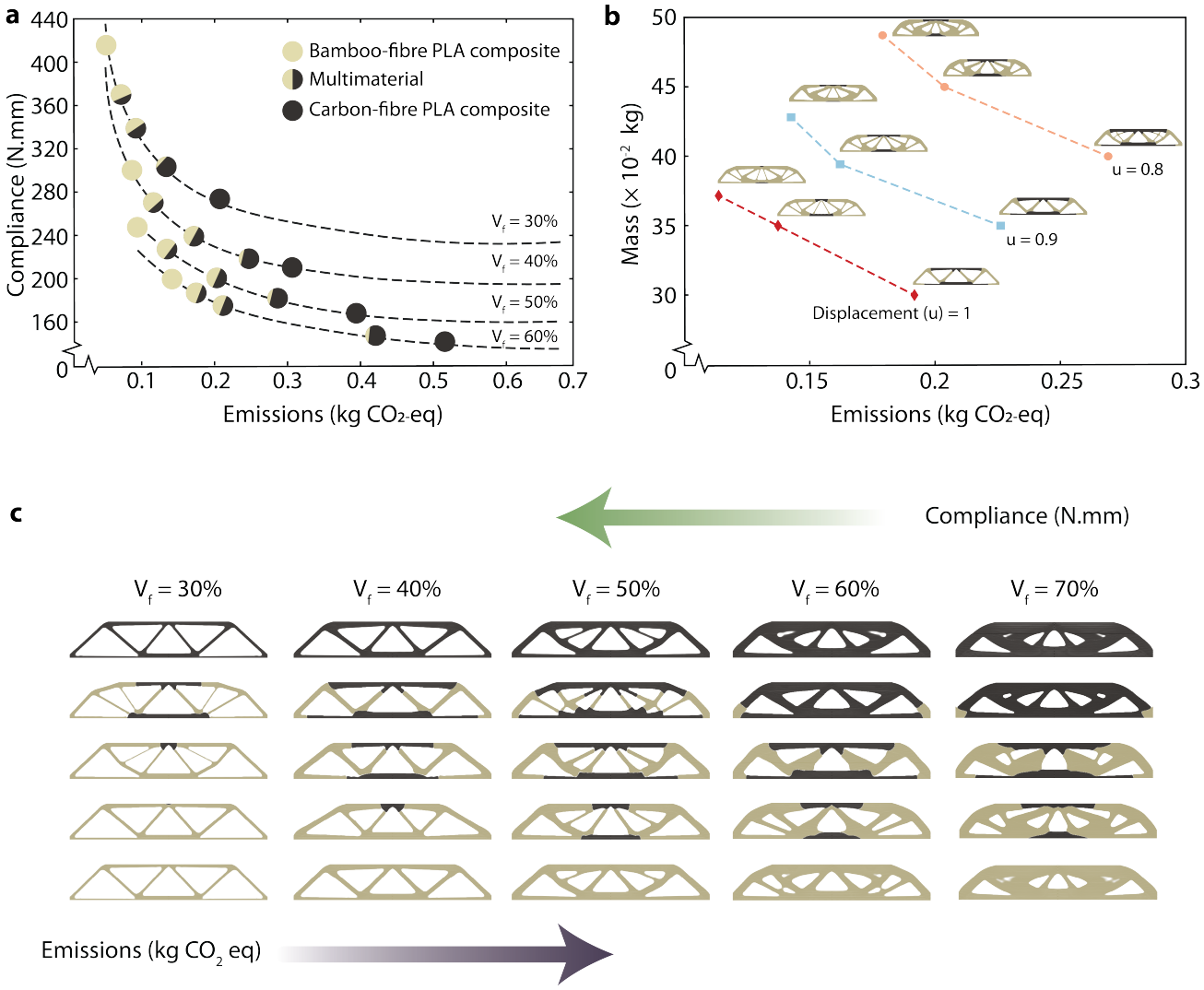

Figure 5: a) Pareto front showing the trade-off between structural compliance and the total volume fraction of the composite. For each prescribed volume fraction (V_f), single-material compliance minimisations were first performed for CF-PLA and Bamboo-PLA to establish the upper and lower bounds. Three intermediate compliance targets were then selected as constraints for the multimaterial CO_2-minimising optimisation, generating hybrid structures with different CF-PLA/Bamboo-PLA proportions. b) Achievable design space obtained by jointly varying the maximum displacement constraint ($u_{\max}$ = 0.8, 0.9, 1.0 mm) and the allowable mass (0.03–0.05 kg in increments of 0.005 kg). Each point corresponds to a CO_2-minimizing multimaterial layout that satisfies, or closely approaches, the imposed constraints. c) Catalogue of optimised multimaterial topologies across volume fractions, showing how CF-PLA concentrates along primary load paths while Bamboo-PLA fills low-stress regions. Together, these results map the feasible combinations of stiffness, mass, and environmental impact, illustrating how designers can select a topology that balances displacement limits, weight budgets, and CO_2 emissions.

map depends on how CO_2 emissions are defined. For example, if use-phase emissions were included for a flying vehicle, the total environmental impact would be governed primarily by structural mass, whereas the present analysis considers only cradle-to-gate embodied emissions.

Conclusion

This work demonstrates that environmental impact can be integrated directly into the design of fibre-reinforced structures through a multimaterial topology optimisation framework in which cradle-to-gate CO_2 intensity acts as an explicit design driver. By allowing carbon-fibre PLA and bamboo-PLA to vary independently and by introducing a continuous fibre-orientation field, the method identifies where high-stiffness reinforcement is structurally essential and where lower-impact material can be deployed without compromising mechanical requirements. The results confirm that optimised fibre alignment can be transferred reliably into printed toolpaths, and that the mechanical response of the printed parts closely follows numerical predictions. This establishes that orientation-controlled multimaterial printing can preserve the mechanical intent defined during optimisation.

Beyond individual optimised designs, the resulting catalogue of solutions constitutes a design map that enables informed, application-specific decision making. By systematically exploring the trade space between stiffness, mass and embodied carbon, the framework provides designers with a continuum of feasible multimaterial layouts, ranging from bamboo-dominated, low-carbon configurations to carbon-rich, high-performance solutions. Depending on functional requirements, such as allowable deformation, weight limits, or sustainability targets, designers can select an appropriate balance between carbon and bamboo reinforcement rather than committing to a single material choice a priori. In this sense, the optimisation framework evolves from a purely generative tool into a quantitative guide for sustainable material allocation.

The main value of this approach lies in converting environmental performance into a controllable and transparent design parameter within a unified workflow that links optimisation, toolpath generation and fabrication. Current limitations include the assumption of linear elasticity and the absence of fully print-aware constraints within the optimisation loop. Nevertheless, the formulation is general and can be extended to incorporate strength, buckling, fatigue, cost, and broader life-cycle indicators. Extending the framework to multi-layer and fully three-dimensional fibre architectures represents a promising direction for future work, opening new pathways for the design of complex three-dimensional structures in which material distribution, fibre orientation and environmental performance are controlled simultaneously. Overall, this work provides a practical foundation for the systematic, application-driven design of low-carbon, high-performance composite structures enabled by multimaterial additive manufacturing.

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Author Contributions

Conceptualisation, Methodology: M.A., G.A., C.J, K.M.; Software: M.A., G.A.; Investigation: M.A., G.A., C.J, A.L, K.M.; Validation: All, Supervision: J.M., K.M.; Writing – original draft: M.A. ; Writing – review & editing: All

Competing interests

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. Source codes will be made available through our GitLab page.

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Supplementary Information: Design and Three-dimensional Printing of Eco-efficient Multimaterial Structures

Design and Three-dimensional Printing of Eco-efficient Multimaterial Structures

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Contents

- Materials and Methods
- Figure SI1. Three-point bending test setup and camera view

Materials and Methods

Material Properties

Two transversely isotropic composite phases were considered: carbon-fibre reinforced PLA (CF-PLA) and bamboo-fibre reinforced PLA (Bamboo-PLA). The longitudinal direction (ℓ) corresponds to the local fibre direction defined by the orientation field, while t denotes the transverse direction. The properties listed in Table 1 were used for constitutive modelling, mass calculation and environmental assessment. The elastic moduli of the composite filaments were estimated from constituent data using standard composite micromechanics. Specifically, the fibre and matrix weight fractions provided by the filament suppliers were converted into volume fractions using the corresponding densities, and the longitudinal modulus was then obtained using the iso-strain rule of mixtures. The transverse modulus, shear modulus and Poisson's ratio were assigned based on representative values for short-fibre reinforced PLA systems reported in the literature and manufacturer datasheets [39]. Material densities were taken from supplier data and used both for mass estimation and construction of the density-dependent terms in the optimisation. The CO_2 intensities correspond to cradle-to-gate greenhouse gas emissions per unit mass. These values were obtained from Idemat and Ecoinvent, and eco-costs midpoint tables [38].

Sensitivity Analysis

For completeness and broader usability, the expressions originally defined for two materials are here extended to an arbitrary number of materials M, enabling multimaterial optimisation within a unified framework.

Table 1: Orthotropic material properties. Subscripts ℓ and t refer to longitudinal and transverse directions, respectively.

Material	E_ℓ (MPa)	E_t (MPa)	$\nu_{\ell t}$	$G_{\ell t}$ (MPa)	ρ (g/mm^3)	CO_2 (kg CO_2-eq/kg)
CF-PLA	16700	3700	0.33	1400	1.27e-3	10.1
Bamboo-PLA	10800	4600	0.36	1700	1.12e-3	2.9

Filtering

Standard density topology optimisation suffers from numerical issues such as checker-board patterns and mesh dependency. In order to overcome these issues, topology optimisation methods are often equipped with filters, explicit constraints, or regularisation to smoothen the density field [40, 41, 31]. Similarly, filtering of orientation variables is also critical to ensure continuous and long fibre paths instead of short and discontinuous paths in post-processing.

In our formulation, we first utilise the convolution filters for density variables $(\rho_e^1, \ldots, \rho_e^M)$ and for the fibre orientation variables (θ_e). The convolution filter refines the elemental variables by weighted averaging them in the mesh-independent neighbourhood of the element. These convolution filters avoid sudden variations in the densities and fibre-orientations in the neighbouring elements. After applying the convolution filters, the filtered variables, denoted as $\tilde{\rho}_e$ and $\tilde{\theta}_e$, and their corresponding sensitivities are given as,

$$\tilde{\rho}_e^m = \frac{\sum_{i=1}^n w_i^e \rho_i^m}{\sum_{i=1}^n w_i} \quad \text{and} \quad \frac{d\tilde{\rho}_e^m}{d\rho_j^m} = \frac{w_j^e}{\sum_{i=1}^n w_i}, \quad m = 1, \ldots, M, \tag{6}$$

$$\tilde{\theta}_e = \frac{\sum_{i=1}^n W_i^e \theta_i}{\sum_{i=1}^n w_i} \quad \text{and} \quad \frac{d\tilde{\theta}_e}{d\theta_j} = \frac{W_j^e}{\sum_{i=1}^n W_i}, \tag{7}$$

with

$$w_i^e = \max(0, r - dist(e, i)) \quad \text{and} \quad W_i^e = \max(0, R - dist(e, i)), \tag{8}$$

where w_i^e and W_i^e are the weight factors to define the neighbourhood of the e^{th} element for densities and fibre orientations, respectively. Both r and R are radius of filters.

For densities, we also apply an additional - Heaviside projection filter [30, 42] on top of convolution filter as,

$$\hat{\rho}_e^m = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\rho}_e^m - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))} \quad \text{and} \tag{9}$$

$$\frac{d\hat{\rho}_e^m}{d\tilde{\rho}_e^m} = \frac{\beta \cdot \operatorname{sech}^2(\beta(\tilde{\rho}_e^m - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))} \quad \text{with} \quad m = 1, \ldots, M, \tag{10}$$

where β and η parameter control the steepness and the threshold of the Heaviside function. We assigned η as 0.5 for preserving the total volume of the structure. β changes during the optimisation following our continuation method. The objective of utilising this filter is to convert the gray designs produced by the convolution filter into black and white designs.

Material Interpolation Scheme for M materials

A notable advantage of the UMMI-2 scheme is its extensibility to multiple material phases. We therefore generalise the interpolation to M materials, for which the elemental elasticity matrix $\mathbf{C}_e$ is given by:

$$\mathbf{C}_e(\hat{\rho}_e, \tilde{\theta}_e) = \mathbf{C}_{\text{void}} + \sum_{m=1}^{M} (\hat{\rho}_e^m)^p \prod_{\substack{n=1 \\ n\neq m}}^{M} (1 - (\hat{\rho}_e^n)^p) \left(\mathbf{R}(\tilde{\theta}_e)^T \mathbf{C}_m \mathbf{R}(\tilde{\theta}_e) \right) \tag{11}$$

where $\mathbf{C}_{\text{void}}$ and $\mathbf{C}_m$ are the elasticity matrices for void phase and material phase m, respectively. $\mathbf{R}$ is the local 2D rotation matrix to rotate the transversely isotropic elasticity matrix according to the filtered fibre-orientation $\tilde{\theta}_e$.

In addition to the elasticity matrix $\mathbf{C}_e$, we also need its derivative with respect to the filtered design variable to calculated the sensitivities. The derivative can be given as,

$$\begin{aligned} \frac{d\mathbf{C}_e}{d\hat{\rho}_e^m} = {} & p\,(\hat{\rho}_e^m)^{p-1} \prod_{\substack{n=1 \\ n\neq m}}^{M} (1 - (\hat{\rho}_e^n)^p) \left(\mathbf{R}(\tilde{\theta}_e)^T \mathbf{C}_m \mathbf{R}(\tilde{\theta}_e) \right) \\ & + \sum_{\substack{n=1 \\ n\neq m}}^{M} (\hat{\rho}_e^n)^p (-p\,(\hat{\rho}_e^m)^{p-1}) \prod_{\substack{r=1 \\ r\neq m \\ r\neq n}}^{M} (1 - (\hat{\rho}_e^r)^p) \left(\mathbf{R}(\tilde{\theta}_e)^T \mathbf{C}_n \mathbf{R}(\tilde{\theta}_e) \right) \quad \text{with} \quad m = 1, \ldots, M \end{aligned} \tag{12}$$

$$\frac{d\mathbf{C}_e}{d\tilde{\theta}_e} = \sum_{m=1}^{M} (\hat{\rho}_e^m)^p \prod_{\substack{n=1 \\ n\neq m}}^{M} (1 - (\hat{\rho}_e^n)^p) \left(\frac{d\mathbf{R}}{d\tilde{\theta}_e}^T \mathbf{C}_m \mathbf{R} + \mathbf{R}^T \mathbf{C}_m \frac{d\mathbf{R}}{d\tilde{\theta}_e} \right) \tag{13}$$

0.1. Environmental impact evaluation

We formulate the environmental objective for an arbitrary number of materials M using a cradle-to-gate assessment, yielding:

$$CO_2(\mathbf{X}) = \sum_{m=1}^{M} \sum_{e=1}^{n} \hat{\rho}_e^m \, CO_2^m \, v_e \tag{14}$$

with the sensitivities,

$$\frac{dCO_2}{d\hat{\rho}_e^m} = CO_2^m \, v_e \quad \text{with} \quad m = 1, \ldots, M \tag{15}$$

$$\frac{dCO_2}{d\tilde{\theta}_e} = 0 \tag{16}$$

where CO_2^m is the CO_2 footprint associated with the unit volume of material phase m.

Toolpath generation

To translate the optimiser's fibre-orientation field into manufacturable deposition paths, custom streamlines were generated using Slith3r, an open-source field-aligned toolpath generator developed at Shaping Matter Lab [43]. Slith3r implements and extends the field-aligned printing methodology introduced by Houriet et al. [22], in which an angle field defined over the domain guides the construction of distance-aware streamlines. Streamlines are inserted iteratively such that the maximum spacing between neighbouring paths does not exceed a prescribed upper bound, while paths closer than a lower bound are trimmed, yielding continuous, non-intersecting curves that satisfy constraints on spacing, curvature, and manufacturability.

To enable multimaterial printing, the optimiser's multimaterial solution was converted into two disjoint material regions within the structural domain Ω, corresponding to CF-PLA and Bamboo-PLA,

$$R_{\mathrm{CF}} = \{\, x \in \Omega \mid \mathrm{material}(x) = \text{CF-PLA} \,\}, \qquad R_{\mathrm{B}} = \{\, x \in \Omega \mid \mathrm{material}(x) = \text{Bamboo-PLA} \,\}.$$

Let $\mathcal{P}$ denote the set of all field-aligned streamlines generated from the orientation field. Material-specific toolpaths are then obtained by intersecting $\mathcal{P}$ with the two regions,

$$\mathcal{P}_{\mathrm{CF}} = \mathcal{P} \cap R_{\mathrm{CF}}, \qquad \mathcal{P}_{\mathrm{B}} = \mathcal{P} \cap R_{\mathrm{B}}.$$

In practice, each region R_{CF} and R_{B} is rasterised into a binary mask and sampled along the streamlines: points falling outside the mask are discarded and segments are split at mask transitions. This procedure produces two disjoint streamline sets conforming to the optimiser's material layout. Slith3r then assigns material-specific line widths and extrusion parameters, producing two valid G-code sequences for CF-PLA and Bamboo-PLA.

Finally, a toolpath-merging pipeline integrates the two G-code files into a single build sequence. Merging assigns extruder IDs, inserts tool-change commands, synchronises travel moves, and ensures clean material transitions without over-extrusion or nozzle collisions. The resulting multimaterial G-code encodes the optimiser's material distribution, the stress-aligned fibre trajectories, and all switching instructions for the Prusa XL multi-toolhead system.

3D printing

PolyLite™ CF-PLA from Polymaker containing 8% milled carbon fiber and Bamboo-PLA from Real Filament, containing 20% bamboo fiber by volume were used as feedstock materials. Prior to printing and between prints, the filaments were dried overnight at 60 °C in a vacuum oven. Samples were fabricated on a Prusa XL printer equipped with five independent print heads, with head 1 dedicated to CF-PLA and head 2 to Bamboo-PLA. Printing was performed with a 0.2 mm layer height, nozzle temperature of 230 °C, bed temperature of 60 °C, and feed rates of 2400 mm/min for the first layer and 6000 mm/min

for subsequent layers. Retraction and deretraction were enabled to minimise stringing, and nozzle purging was carried out at the build plate edge between layers. Five identical parts were printed simultaneously to ensure uniform processing conditions.

Three-point bending

Three-point bending tests were performed to characterise the flexural response of the multimaterial printed beams. All experiments were carried out using a servo-hydraulic universal testing machine (Zwick-Roell, Germany) equipped with a calibrated 20 kN load cell. A dedicated bending fixture with a support span of 19.8 cm was used, and both the loading nose and the two support rollers were machined with 5 mm radii to minimise local stress concentrations.

Printed specimens had a nominal thickness of 3.6 mm and width of 7.5 mm. Tests were conducted under displacement control at a crosshead speed of 1 mm/min, with force and machine displacement recorded at ≥50 Hz to capture the elastic response with sufficient resolution. The bending setup ensured consistent orientation of the streamline-aligned toolpaths relative to the loading axis across all specimens.

Nominal flexural stress σ and mid-span strain ε were computed from the measured load P and compliance-corrected mid-span deflection δ using the standard three-point bending relations:

$$\sigma = \frac{3PL}{2bt^2}, \tag{17}$$

$$\varepsilon = \frac{6\delta t}{L^2}, \tag{18}$$

where L is the support span, b the specimen width, and t the specimen thickness. Because the printed beams were topology-optimised, materially heterogeneous, and non-prismatic, these expressions provide nominal outer-fibre stress and strain rather than a true homogeneous-beam response. Consequently, the classical Euler-Bernoulli definition of flexural modulus is not applicable. Instead, following established practice in architected and metamaterial structures, we report an *effective bending stiffness*, expressed as an *effective flexural modulus* E_{eff}, defined directly from the measured structural response as:

$$E_{\text{eff}} = \left(\frac{d\sigma}{d\varepsilon}\right)_{\text{elastic}}, \tag{19}$$

where the derivative is obtained from a least-squares linear regression over the initial linear elastic segment of the stress–strain curve. To ensure consistent comparison across multimaterial configurations, the regression window was restricted to the elastic-dominant region where $\sigma \leq 200$ MPa, beyond which some Bamboo-PLA–rich specimens exhibited early nonlinearity.

Mechanical data were processed using baseline correction, light Savitzky–Golay filtering, and interpolation onto a common strain grid for specimen-to-specimen comparison. Outlier curves were identified based on their deviation from group-average behaviour, and representative curves were used to compute flexural modulus and 95% confidence intervals.

DIC

A two-camera digital image correlation (DIC) system was used to obtain full-field strain and displacement measurements during the three-point bending tests. Specimen surfaces were prepared with a high-contrast stochastic speckle pattern, and images were recorded at 2 Hz using 9 MP cameras equipped with 50 mm lenses. The cameras were calibrated prior to testing to ensure accurate stereoscopic reconstruction of the displacement field.

DIC analysis was performed using Vic-3D (version 8), with a subset size of 20 pixels and a step size of 7 pixels, providing a balance between spatial resolution and noise suppression. The software tracked the deformation of the speckle pattern throughout the bending process, yielding full-field strain maps and precise mid-span displacement values.

Three-point bending test setup and camera view

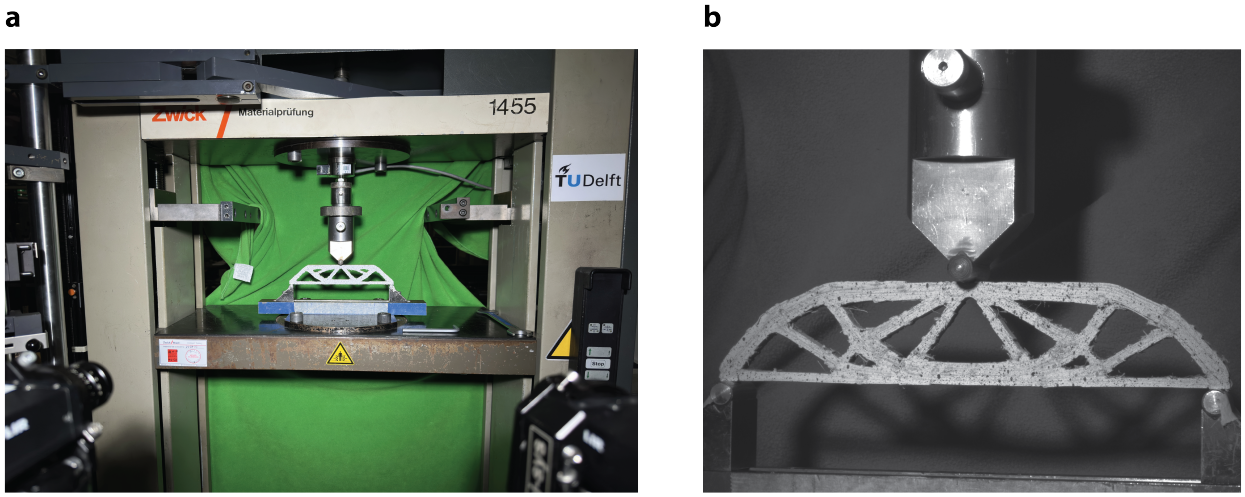

Figure SI1: Three-point bending testing configuration for the printed multimaterial MBB beams. a. Experimental setup in the universal testing machine with a three-point bending fixture. b. Image acquired by the camera during the test, used for deformation and strain evaluation.